MCO-1D PRE-FLIGHT CHECKLIST

Rev 20 · Run before every flight — do not skip

1. PHYSICAL INSPECTION

- ■ **All 8 wings at 60° sweep, no binding or interference**
- ■ All 16 servo linkages fully seated and locked
- ■ **All 8 ropes seated in groove — no fraying visible**
- ■ 8 encoder magnets secure on motor shafts — no wobble
- ■ CG balance check: aircraft hangs level ±5°
- ■ No loose screws, bolts, or wire connectors
- ■ **Flight battery secured, connector fully locked**
- ■ **Servo LiPo: charged, above 7.2V**

2. ELECTRONICS

- ■ 8×ODrive S1 power lights: solid green
- ■ **Pixhawk GPS: solid blue LED, ≥6 satellites**
- ■ Pitot tube: unobstructed, facing forward, no kinks
- ■ RC receiver: binding confirmed (solid LED)
- ■ **RC range check: all sticks respond correctly in GS**
- ■ **Lever convention: forward = more thrust confirmed**

3. SOFTWARE (GS COMMANDS)

- ■ **Pi booted: mco_main.py running**
- ■ GS laptop connected to Pi port 5757
- ■ **persist_load_all: learning restored — confirm output**
- ■ soft_stop_status: all 8 wings PASS
- ■ cg_status: stability margin positive
- ■ **failure_briefing: 0 Catastrophic**
- ■ **fault_summary: no active faults**
- ■ **tick_error_report: no unresolved errors**
- ■ gps_ok=True, pitot valid in telemetry
- ■ **surface_status: correct region for this flight**
- ■ inbox_check: no pending knowledge imports

4. BEFORE ARM

- ■ **Area clear: 10m radius, observers behind pilot**
- ■ Wind below 8 m/s (storm_char_report if higher)
- ■ **Home position set at ARM — do not move after**
- ■ **Emergency procedures briefed to all persons**
- ■ AI GS: ws://localhost:5758 connected

ARM: type `arm` in GS or AI

MCO-1D IN-FLIGHT EMERGENCY CARD

Keep visible during all flights

WING FAILURE — PROGRESSIVE

- 6–7 wings: degraded warning, aircraft continues
- 5 wings: advisory to land — pilot may continue
- **≤4 wings: LAND NOW auto-engages immediately**
- Lever sets descent rate (50%=0.20m/s, 100%=0.08m/s)
- **Pilot RETAINS full roll/yaw/pitch authority throughout**
- **Steer to clear area — do not fight the descent**

RC LINK LOSS

- Aircraft climbs to 30m, orbits, waits 10s
- **Link not restored: RTH to ARM GPS position**
- **RTH descends 0.3m/s — clear landing zone**

GPS LOST / SPOOFED

- **GS: gps_status — check level and DR confidence**
- Dead reckoning continues for ~60s before degrading
- Rescue vessel can inject fix: accept_external_fix
- Transit holds last known bearing in GPS-denied mode

STALL WARNING

- stall_detector_status > 0.45
- **Increase lever — do NOT reduce**
- Reduce bank, allow nose to drop

IMMEDIATE ACTIONS — ANY EMERGENCY

- **Call CLEAR — warn all observers**
- **Maintain visual contact at all times**
- Lever to 80%+ unless instructed otherwise

EMERGENCY STOP — LAST RESORT ONLY

- **GS: emergency_stop OR panel: RED E-STOP**
- **All motors stop — aircraft drops immediately**
- **ONLY if crash certain AND people at risk**

ABORT FLAP-GLIDE IF SINKING

- **flap_glide_disable — returns to continuous flapping**
- All pairs restart within one stroke cycle
- **Lever up to arrest descent rate first**

LAND_NOW FROM THE GROUND

- Wing frequency audibly reduces near ground
- Descent: 0.08–0.50 m/s depending on lever
- May drift with wind — steer gently
- **Full pilot authority throughout**
- **After landing: lever 0%, then GS: disarm**

KEY COMMANDS

- fault_summary / tick_error_report
- gps_status / flap_glide_disable
- wig_floor_disable (deliberate ditching only)

MCO-1D POST-FLIGHT LEARNING CARD

Rev 20 · Run after every flight — do not skip

STEP 1 — SAFE FIRST

- ■ Lever to 0%
- ■ **GS: disarm**
- ■ **Confirm all wings stopped and still**
- ■ Disconnect flight battery
- ■ Fold wings for storage

STEP 2 — SAVE ALL LEARNING (BEFORE POWER OFF)

- ■ **GS or AI: persist_checkpoint**
- ■ Wait: "Learning persisted — N components"
- ■ Confirms: wave library + ME archive + strategy genomes
- ■ **GS: knowledge_export_now**
- ■ **Copy outbox to USB or laptop**

STEP 3 — RECORD OBSERVATIONS

- ■ GS: gs_inject_note "your observation"
- ■ Have an idea for a mechanism combination?
- ■ **sg_inject_strategy task=X mech_a=0.6 mech_b=0.4**
- ■ Have a kinematic idea? me_inject_operator_idea
- ■ **Storm was rough? Check storm_char_report**

STEP 4 — REVIEW LEARNING

- ■ **GS: learning_history**
- ■ GS: mechanism_weight_report — which mechanisms rank highest
- ■ GS: me_archive_status — new MAP-Elites niche champions?
- ■ GS: sg_archive_status — new strategy combinations?
- ■ GS: hypothesis_results — hypotheses promoted
- ■ **GS: tick_error_report — any subsystem errors?**

STEP 5 — FLEET SHARING

- ■ Each drone: knowledge_export_now after flight
- ■ **merge_fleet_knowledge d1.json d2.json**
- ■ **knowledge_import merged.json on each drone**
- ■ Also: me_archive_import fleet_me.json (MAP-Elites)

STEP 6 — INJECT IDEAS FOR NEXT FLIGHT

- ■ **Kinematic: me_inject_operator_idea name=X param=val**
- ■ **Mechanism combo: sg_inject_strategy task=X ...**
- ■ Scoring focus: focus for next session
- ■ Or tell AI assistant in plain English

SHUTDOWN: power off Pi ONLY AFTER persist_checkpoint

MCO-1D ASSEMBLY SEQUENCE CARD

Rev 20 · First build — follow in order

PHASE A — SERVOS (DO FIRST)

- ■ **Buy: 18×AGFRC IB53BHU + AGF-SPV3 programmer**
- ■ **Programme ALL 16 flight servos to 170° max travel**
- ■ ADC saturates at 180° — 170° gives 10° margin
- ■ Label: W1p,W1s,W2p…W4s
- ■ Install servos — do not connect ropes yet

PHASE B — DRIVES AND ENCODERS

- ■ **Mount 8×ODrive S1 (node_id 0–7)**
- ■ CAN-FD bus: daisy-chain all 8, 120Ω terminators
- ■ **Install encoder magnet on each motor shaft**
- ■ **Verify: position reads full arc smoothly**
- ■ **Run mco_odrive_setup.py — set sector arc limits**
- ■ GS: soft_stop_status — all 8 PASS

PHASE C — ROPES (SAFETY CRITICAL)

- ■ Thread rope through each sector wheel groove
- ■ **FATIGUE TEST: drive at rated current to failure**
- ■ **Record I²×cycles — update ROPE_LIFE_I2_CYCLES**
- ■ **F-602: include shock loading component (see Addendum G)**
- ■ Connect ropes, check tension
- ■ **No fraying at reversals under power**

PHASE D — MEASUREMENTS

- ■ **Run python3 mco_build_measurements.py**
- ■ **A1: actual MTOW — weigh complete aircraft**
- ■ C1: gear ratio / C2,C3: sector arc angles
- ■ **K_PITCH_MM_PER_DEG: tilt 5°, measure CG shift**

PHASE E — ELECTRONICS AND RC

- ■ Pixhawk: GPS antenna clear sky view
- ■ **Pitot: forward-facing, no bends**
- ■ Wire MCP3208 ADC for servo feedback
- ■ **Run mco_rc_verify.py — confirm all channels**
- ■ **Lever: forward = more power (FW convention)**
- ■ Rudder: right pedal = right yaw

PHASE F — SOFTWARE COMMISSIONING

- ■ **Boot Pi: mco_main.py, no errors**
- ■ Laptop: python3 mco_ws_bridge.py [pi_ip]
- ■ AI ground station in browser
- ■ **TC-01 bench: tc01_start**
- ■ **TC-02 captive hover: tc02_start**
- ■ **TC-03 truck FW (validates FG_MIN_GLIDE_V_MS): tc03_start**
- ■ **TC-04 first free flight: tc04_start**

MCO-1D SAR OPERATIONS CARD

Rev 20 · Search and Rescue mission procedures

TRANSIT TO SCENE

- ■ Climb 30m+, lever to 70%
- ■ Airspeed hold engages after 1.5s stable lever
- ■ **Or: transit_to lat lon alt_m for autopilot transit**
- ■ Autopilot selects strategy based on wind conditions
- ■ transit_status shows distance, bearing, crab angle
- ■ **Take stick > 5% at any time to override**

APPROACH AND SITE HOLD

- ■ Reduce lever: blend zone 3–5 m/s
- ■ **Below 3 m/s: hover auto-engages**
- ■ **Crew hover ON: stick = velocity not attitude**
- ■ **Release stick: decelerates, stops**
- ■ **GS or AI: position_hold_engage to lock GPS**

GPS DEGRADED AT SCENE

- ■ **gps_status shows level and DR confidence**
- ■ Position hold: heading + altitude hold mode
- ■ Rescue vessel can provide fix: accept_external_fix
- ■ **Confidence < 10%: use visual piloting only**

LOITER / ENDURANCE

- ■ **h36_engage for minimum power draw**
- ■ focus endurance for optimizer assistance
- ■ h36_loiter_report: time remaining
- ■ **RTH with minimum 20% battery reserve**

CREW HOVER

- ■ **Default ON in hover — no switch needed**
- ■ **Stick centred = 0 m/s = holds position**
- ■ Small stick = slow creep; large = 5m/s fwd
- ■ Disable: crew_hover_disable (classic feel)

ROUGH WEATHER OPERATIONS

- ■ **focus storm_resilience if Beaufort > 6**
- ■ storm_char_report shows current wind limits
- ■ Attitude hold compensates toward commanded attitude
- ■ **Load shedding auto-activates above 2.0g**
- ■ W1 pair stops automatically — 3-pair flight continues

RETURN TO HOME

- ■ **AI GS: "return to home"**
- ■ Climbs, tracks to ARM GPS position
- ■ **Descends 0.3m/s — clear landing zone**
- ■ **Take stick to override at any time**